placeholder

Anonymous Author(s)
Affiliation
Address
email

Abstract

placeholder

1 Introduction

[This is only a skeleton of the intro] Much of the world knowledge can be expressed in terms of relations, but oftentimes such relations are graded, implied, and nonliteral. Such kind of implicit inference of event relations is common in human language processing... (cite) Talk about eliciture and why its implied relationships between any two events are non-mandated. (cite)

Previous studies have attempted to show how LMs can build a world model through pragmatic inference... (cite) To the best of our knowledge, Pan & Kehler is the only work that examines LM's ability to perform eliciture inference. They demonstrate that LMs not only draw eliciture but also leverage eliciture to guide downstream sentential processing. we aim to extend on their findings by evaluating models of different families and studying the internal representations of implicit causal inference.

The benefits of understanding elicitors in LMs are multifold: 1. whether LMs–trained with distributional objective–demonstrate human-like nonliteral inference in non-mandated setting 2. more comprehensive views into how LMs build an internal world model 3. Potentially understand the implicit bias in LMs, which has implications in safety and alignment

To these aims, we explore three main questions:

RQ1 *Do model behaviors reflect inferences about implicit causal relationships within sentences?*
RQ2 *Do LMs form internal representations of implicit causal relationships?*
RQ3 *If **RQ2** is true, do representations of implicit causality generalize to naturalistic settings with potential real-world implications?*

Our contribution is...

2 Background

3 Experiments

3.1 Measuring Implicit Causal Inference through Model Behaviors

Models. In this and all of the following experiments, we test Gemma 2 2B/9B [Gemma Team, 2024], Qwen3 4B/8B [Qwen Team, 2025], Llama 3.2 3B [Meta AI, 2024], and Llama 3 8B [Dubey et al., 2024] base checkpoints.

Dataset. We use 150 manually authored pairs of sentences. In each pair, the modifier or relative clause in the critical sentence provides a plausible cause of that outcome based on physical or everyday world knowledge; in the *control* sentence, it provides an incidental detail. The dataset generally follows the format of [person] [modifier content] [outcome]. For example, *The executive who falsified*

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

quarterly revenue figures was fired is paired with *The executive who wore a blue silk tie was fired*. This dataset extends the no-context eliciture paradigm of Pan and Kehler [2025], which tests whether an apparently descriptive modifier is nevertheless understood as an explanation.

Yes/No. We first present the sentences of [person] [modifier content] [outcome] as context and then ask, "Does [modifier content] explain why [outcome]?" We compute the probabilities of the complete answers *Yes* and *No*, including capitalization variants, and normalize them to obtain $P(\text{Yes})$.

7-point Likert. We present the tested sentences as context and then ask how strongly the [modifier content] explains the [outcome] on a scale from 1 (definitely not) through 4 (uncertain) to 7 (definitely yes). We compute a probability for each of the seven complete answers and use their probability-weighted mean as the model's rating.

Surprisal. Following Pan and Kehler [2025], we remove the sentence-final period and append *", and I don't know why."* We use the surprisal of this continuation as the score. The intuition is that if the modifier has already supplied an explanation, subsequently claiming *I don't know why* should be less expected and therefore more surprising.

For all three measures, paired accuracy is the proportion of pairs in which the critical sentence scores strictly above its control; chance is .5, ties count as incorrect.

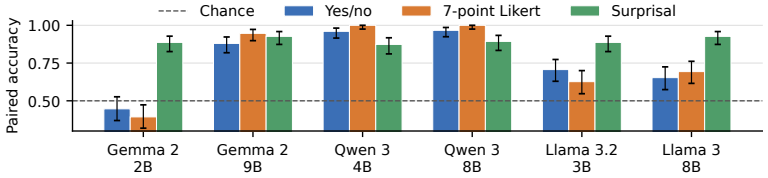


Figure 1: Paired accuracy of model judgments on implicit causal inference items, comparing Yes/No prompting, 7-point likert rating, and the surprisal of the continuation *", and I don't know why."* Higher accuracy indicates better performance in implicit causality inferences.

Results. As shown in figure 1, models demonstrate implicit causal inference most consistently through surprisals. Not all models can faithfully report their implicit causal inference through Yes/No and 7-Point Likert rating, and the performance of such metalinguistic causal inference increases with model size.

3.2 Identifying Internal Representations of Implicit Causality Inference

Given that LMs behaviorally demonstrate implicit causal inference, do LMs form internal representations of implicit causality? Specifically, we ask two questions:

1) Do model activations distinguish implicit causal relationships from other relations? If a representation is specific to causality, a causal direction should separate causal pairs well but transfer less reliably to temporal or plausibility pairs.

2) If such representations exist, could the representations of cross-sentence implicit causality transfer to within-sentence causality (eliciture)? Within-sentence causal inferences are less strong and more optional compared to cross-sentence causal inferences. If the representations of implicit causality inference is robutst, it should transfer across syntactic variations.

Datasets. We use three datasets of matched critical/control pairs. The training and in-distribution evaluation sets do not share items. The *causal* dataset contains 384 training and 96 evaluation pairs. In each pair, the second sentence in the critical item explains the first sentence; in the control item, it provides an incidental detail. For example, *Anika avoided the Amber Cafe. The Amber Cafe repeatedly mixed up Anika's allergy order* is paired with *Anika avoided the Amber Cafe. The Amber Cafe had a copper sign*. The *temporal* dataset contains 800 training and 200 evaluation pairs. The critical item pairs an occupation with an event that is likely during that occupation, whereas the control pairs the same event with a less compatible occupation (e.g., a lifeguard versus a translator getting a sunburn). This dataset tests occasion compatibility, not temporal order. The *plausibility* dataset contains 56 training and 14 evaluation pairs. The critical item describes a probable event, whereas the control describes a physically impossible event (e.g., baking a cake in an oven versus

a freezer) [Hu et al., 2025, Lepori et al., 2026]. For Figure 3, we use the 150 within-sentence critical/control pairs from Section 3.1. These pairs are held out from CAA training.

CAA directions. Contrastive Activation Addition (CAA) represents a relation with a direction in a model's activation space [Rimsky et al., 2024]. At each tested layer, we subtract the final-token activation of every control item from that of its matched critical item, then average these differences over the training pairs. An evaluation item receives a scalar score by projection onto this direction of averaged difference; paired accuracy is the proportion of evaluation pairs for which the critical item scores higher.

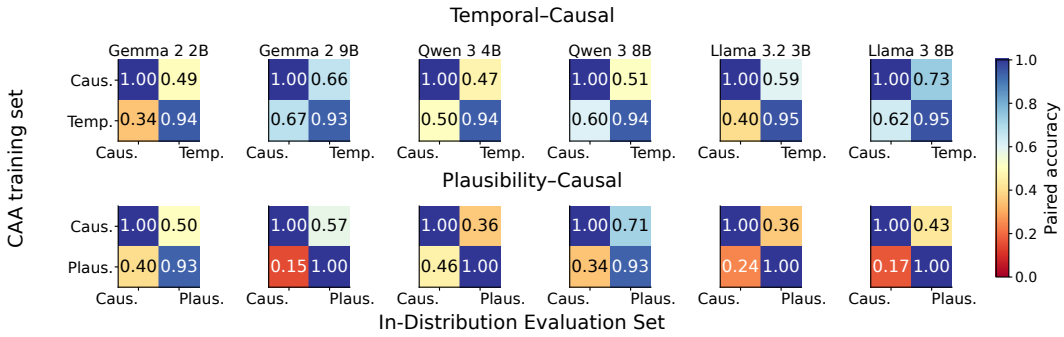


Figure 2: Cross-transfer paired accuracy between causal and temporal relations (top row) and causal and plausibility relations (bottom row). Within each heatmap, rows show the CAA training set and columns show the in-distribution evaluation set. **Causal CAA directions are largely relation-selective and distinct from temporal and plausibility directions.**

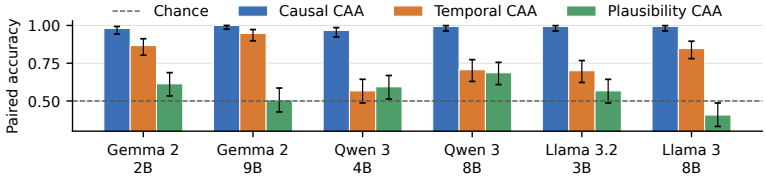


Figure 3: Paired accuracy of causal, temporal, and plausibility CAA directions on 150 unseen within-sentence causal inference pairs with 95% CIs. **Causal CAA directions generalize consistently to within-sentence implicit causal inference.** Temporal and plausibility CAA directions transfer less consistently to within-sentence implicit causal items.

Results.

3.3 Generalization to Naturalistic Scenarios

Dataset.

Table 1: Mean paired differences in causal CAA projection in naturalistic causal scenarios. Positive values indicate that a demographic group has a greater mean projection than the white-male condition. Bolded values indicate statsitically significant difference from the white-male projection scores. **Demographic descriptions are associated with differences in causal CAA projections.** Minority females generally have larger projection raw mean scores than minority males along the implicit causal directions.

	Gemma 2 2B	Gemma 2 9B	Qwen3 4B	Qwen3 8B	Llama 3.2 3B	Llama 3 8B
Minority male − White male	**.41 ± .32**	**1.9 ± .40**	.073 ± .093	**.29 ± .17**	.025 ± .029	**.030 ± .020**
Minority female − White male	**.57 ± .32**	**2.3 ± .41**	**.15 ± .10**	**.39 ± .15**	**.069 ± .033**	**.060 ± .021**

Results.

4 Discussion

References

Abhimanyu Dubey et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. URL https://arxiv.org/abs/2407.21783.

Gemma Team. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*, 2024. URL https://arxiv.org/abs/2408.00118.

Jennifer Hu, Felix Sosa, and Tomer Ullman. Shades of zero: Distinguishing impossibility from inconceivability. *Journal of Memory and Language*, 143:104640, 2025.

Michael A. Lepori, Jennifer Hu, Ishita Dasgupta, Roma Patel, Thomas Serre, and Ellie Pavlick. Is this just fantasy? language model representations reflect human judgments of event plausibility. In *The Fourteenth International Conference on Learning Representations*, 2026. URL https://openreview.net/forum?id=Czul60ELOH.

Meta AI. Llama 3.2 model card. Hugging Face model documentation, 2024. URL https://huggingface.co/meta-llama/Llama-3.2-3B.

Dingyi Pan and Andrew Kehler. Pragmatic competence in LLMs: The case of eliciture. In Carolyn Jane Anderson, Frédéric Mailhot, and Grusha Prasad, editors, *Proceedings of the Society for Computation in Linguistics 2025*, pages 388–391. Association for Computational Linguistics, 2025. URL https://aclanthology.org/2025.scil-1.39/.

Qwen Team. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. URL https://arxiv.org/abs/2505.09388.

Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15504–15522. Association for Computational Linguistics, 2024.

A Raw CAA Projection Distributions

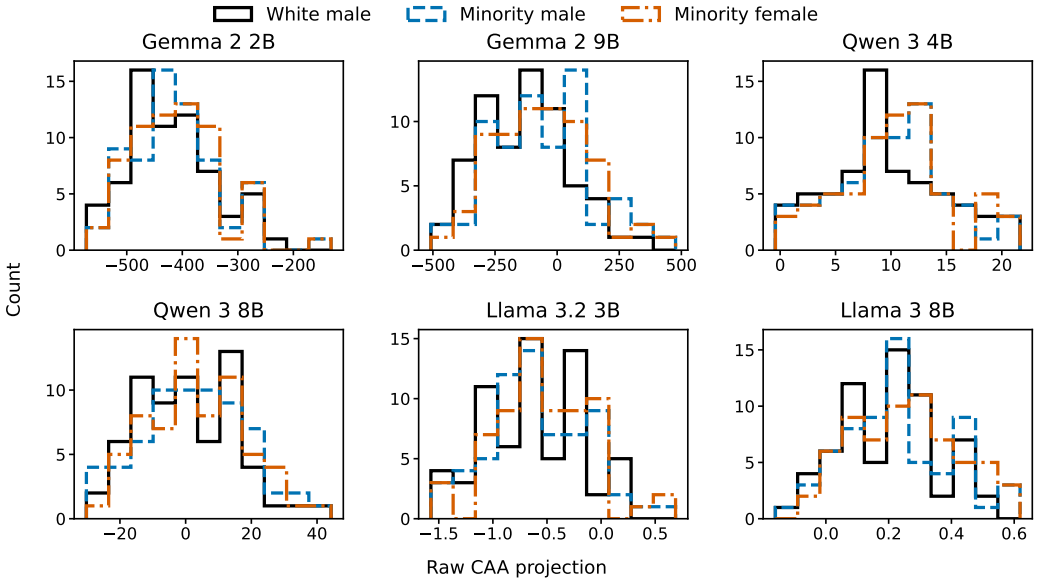


Figure 4: Raw causal CAA projection distributions for three matched demographic conditions across 65 scenarios per model. **The three conditions have substantially overlapping projection distributions in every model.** Colors and line styles distinguish the White-male, minority-male, and minority-female conditions.